MIL-T-27/193A
15 May 1978
SUPERSEDING
MIL-T-27/193
18 October 1977

MILITARY SPECIFICATION SHEET

TRANSFORMERS, POWER, STEPDOWN, TF5S03ZZ and TF5T03ZZ (A)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the transformers described herein shall consist of this document and the latest issue of Specification MIL-T-27

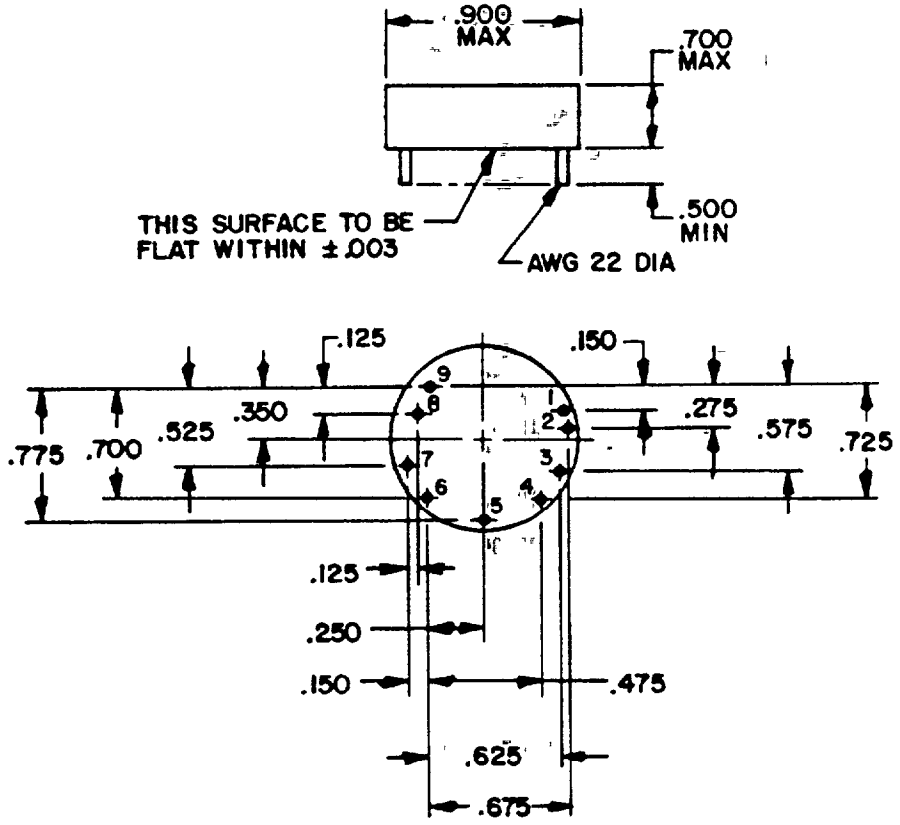
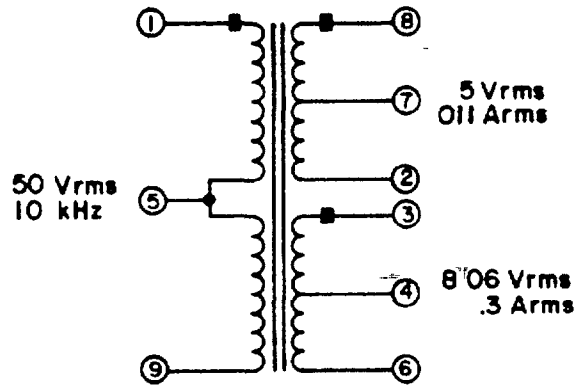


FIGURE 1. Dimensions and configuration.

(A) denotes change

FSC 5950

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WORKING VOLTAGE: 100 V MAX

CIRCUIT DIAGRAM AND MARKING

INCHES	MM	INCHES	MM
003	.08	525	13.34
.125	3.18	.575	14.60
150	3.81	.625	15.88
.250	6.35	.675	17.14
275	6.98	.700	17.78
.350	8.89	.725	18.42
475	12.06	.775	19.68
.500	12.70	.900	22.86

NOTES
1. Dimensions are in inches
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm
3 Unless otherwise specified, tolerance is ±.005 (.13 mm).
4. Marking shall be on the top of the case.

FIGURE 1 Dimensions and configuration - Continued

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REQUIREMENTS: (When numbers in parentheses, i.e. (1-2) are used, they indicate the winding and the extreme terminals of the winding.)

Electrical ratings:
Primary voltage (1-9): 50 volts rms, 10 kilohertz.
Primary current. .006 ampere rms.
Primary power: .3 volt ampere.
Secondary voltage:
(2-8): 5 volts rms, .011 ampere rms.
(3-6): 8.06 volts rms, .3 ampere rms.
Working voltage 100 volts, maximum
DC resistance:
(1-9): 10.6 ohms.
(2-8) 1.5 ohms.
(3-6): .52 ohm

Design and construction:
Dimensions and configuration: See figure 1.
Duty cycle Continuous.
Case: Encapsulated.
Material: Epoxy.
Terminals: Pin type.
Material Type S, tin, lead, or lead alloy coated in accordance with QQ-W-343
Diameter: AWG 22 (.025 inch)
Height: .500 inch minimum.
Weight: .5 ounce.
(A) Operating temperature range: see table I

Terminal strength: MIL-STD-202, method 211, test condition A, 2 pounds.

Dielectric withstanding voltage (at sea level): 300 volts rms.

Electrical characteristics:
No load: With 50 volts rms and 10 kilohertz in (1-9)
Current in (1-9): 10 milliamperes, maximum.
Power in (1-9) .32 watt maximum.
Voltage across (8-2): 5 volts rms ±1 percent.
Voltage across (3-6): 8.06 volts rms ± 1 percent.
Rated load. With 50 volts rms and 10 kilohertz (1-9):
Voltage across (8-3). 5 volts rms ±10 percent.
Voltage across (3-6) 8.06 volts rms ±10 percent.
Polarity: Additive with terminals 9 and 8, and 2 and 3 connected

(A) Temperature rise: 20°C with 50 volts rms, 10 kilohertz across (1-9) at specified ambient temperature. (see table I)
Vibration (high frequency) MIL-STD-202, method 204

Shock (specified pulse) MIL-STD-202, method 213, test condition A.

Marking location See figure 1.

Part number M27/193-(dash number from table I)

(A) TABLE I Temperature characteristics

Dash no	Type designation	Ambient temperature	Temperature rise	Operating temperature range
01	TF5S03ZZ	110°C	20°C	-55° to +130°C
02	TF5T03ZZ	150°C	20°C	-55° to +170°C

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Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy - SH, OS
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - ME, AR, WC
Navy - AS, MC
Air Force - 19

Preparing activity:
Army - EL

Agent:
DLA - ES

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DOCUMENT IDENTIFIER AND TITLE MIL-T-27/193A TRANSFORMERS, POWER, STEPDOWN, TF5S03ZZ AND TF5T03ZZ

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

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B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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☐ YES ☐ NO *(If "Yes", in what way?)*

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